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RESEARCH ARTICLE

Decoding the Molecular Mechanism of Cellar Aroma Loss Between the Traditional and Modern Upper Steaming During the Strong-Aroma Type Baijiu Brewing

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ABSTRACT

To investigate the distinctive cellar aroma differences between traditional and modern upper steaming distilled strong-aroma type of Baijiu (SAB), Napping-ultraflash profile (Napping-UFP) combined with flavoromics was used. Significant distinctions of the cellar-aroma among 18 different upper steamed distilled samples were observed between LZLJ-T7 and LZLJ-M7 using Napping-UFP. Moreover, 44 odorants with FD ≥ 9 were identified using comparative aroma extract dilution analysis and further quantified using multiple quantitative techniques. Among them, 28 odorants with OAVs ≥ 1 were identified as important. In addition, 18 key discriminant mark substances were identified through partial least squares discriminant analysis. Further verification using aroma recombination and omission identified 17 key aroma compounds in LZLJ-T7 and 14 in LZLJ-M7. Partial least squares regression revealed ethyl hexanoate, ethyl 3-phenylpropionate, isovaleric acid, dimethyl trisulfide and ethyl caprylate as the primary contributors to the differential cellar aroma between traditional and modern upper steaming distilled SAB.

1 | Introduction

Strong-aroma type Baijiu (SAB), known as 'Luzhou-flavour Baijiu', accounts for approximately 70% of production and sales in China [1, 2]. This dominance is largely attributed to its unique brewing process, especially the solid-state fermentation in mud pits, which imparts Baijiu with the 'rich cellar aroma, sweet and refreshing taste, harmonious fragrance and long-lasting aftertaste'. Among these, the complex flavour dominated by the 'cellar aroma' has become one of the hallmark characteristics of SAB [3]. Previous research has shown that certain phenolic compounds, sulphur-containing compounds, acids, and esters in Baijiu and pit mud are closely related to cellar aroma [4–7]. Furthermore, 3-methylindole, which gives a mud and animal-like odour, has been determined to be a typical potent odorant responsible for the cellar aroma of the base SAB [2, 8].

Recently, the distillation process of SAB has been demonstrated to significantly influence the generation of cellar aroma and the quality of flavour [9]. For instance, most alcohol-soluble and low molecular weight compounds showed a downward trend with the extension of distillation time, but water-soluble, high molecular weight, and high boiling compounds showed the opposite result. This shift results in the presence of different key aroma compounds in base SAB samples at various distillation stages [10, 11]. *p*-Cresol primarily distils out during the latter distillation

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stage. By carefully controlling timing, excessive distillation of *p*-cresol can be prevented, thereby preserving flavour [12]. With the technological advancement, traditional production technologies of Baijiu are increasingly shifting towards mechanisation [13], adopting mechanical equipment to replace manual operations [14]. Among them, the upper steaming distillation, the key step in Baijiu production [11], has been automatized in large enterprises with systems that precisely control temperature, pressure, and flow to ensure effective separation and extraction of alcohol and aromatic compounds (Figure 1) [15].

Although mechanisation has improved efficiency and consistency, machines cannot fully replicate the precision and nuanced techniques that skilled artisans achieve through manual operations [16]. This leads to the noticeable differences in flavour and quality between Baijiu produced by mechanised and traditional technologies [17]. In recent studies, the flavour compounds and aroma profiles of *Xiaoqu* light aroma type Baijiu produced using modern and traditional technologies were analysed using headspace-solid phase microextraction–gas chromatography–mass spectrometry, gas chromatography–olfactometry–mass spectrometry (GC–O–MS), and sensory evaluation. The results indicated that although the key flavour components are similar between the two methods, there are notable differences in aroma activity values and intensity [18]. Further research on Sesame and Laobaigan aroma types highlighted these differences using gas chromatography with a flame ionisation detector (GC–FID), ultraperformance liquid chromatography, and liquid–liquid microextraction (LLME), along with hexamethyldisilazane derivatization paired with gas chromatography–mass spectrometry (GC–MS). Specifically, it was observed that mechanised Sesame aroma base Baijiu contains higher total amounts of lactones, phenols, acids and sulphur compounds but lower total amounts of sugars, sugar alcohols, esters, furans, alcohols, terpenes and acetals compared with traditionally produced Baijiu [16]. For Laobaigan aroma base Baijiu, mechanisation significantly increased the levels of 4-ethylphenol, 4-ethylguaiacol, trehalose and D-glucose, while the levels of ethyl butyrate, ethyl hexanoate, ethyl octanoate, ethyl decanoate, ethyl lactate, *n*-butanol, butyric acid and hexanoic acid decrease significantly [16]. Apart from that, during the SAB production, mechanised base SAB yields a slightly higher total output than manual methods and produces a larger proportion of premium-grade Baijiu. The aroma enhancement in mechanised SAB is generally superior, particularly in extracting total esters and ethyl hexanoate [19]. Further analysis using GC–FID and GC–MS revealed that acetals significantly contribute to the flavour of traditionally brewed SAB, while isobutyl acetate is crucial in mechanised versions [20]. Although existing studies have explored the flavour differences between mechanised and traditional SAB, there are no reports detailing the differences in characteristic cellar aroma between these two production methods.

To address the aforementioned challenge, recent studies have increasingly used novel sensory descriptive techniques and compared flavour characteristics of a great variety of products [21, 22]. In the research of sensory analysis of Baijiu, descriptive analysis methods are the most commonly used [23]. However, because of the lengthy training for evaluation panels, new rapid descriptive analysis methods have emerged, including Check-All-That-Apply [24], Free Choice Profiling [25], Polarised Sensory Positioning [26], Flash Profiling [27], and Projective Mapping [28]. Time-related testing methods, such as Temporal Check-All-That-Apply [29], Temporal Dominance of Sensations [30], and Time-Intensity [31], were also considered. Napping is a new method derived from Projective Mapping that evaluates the similarities and differences between samples. It first appeared in wine research and is typically combined with ultra-flash profiling (UFP) to obtain the sensory characteristics [32, 33]. Currently, the Napping-UFP method is widely used in the food industry, especially for fruit [34], meat products [35], beverages [36], and dairy products [37]. In the sensory analysis research of Baijiu, only a few studies have applied the Napping-UFP method to distinguish the sensory attributes of Yubingshao Baijiu and rice aroma type Baijiu [38]. However, there are no reported applications of this method in studying the characteristic cellar aroma of SAB.

In this study, 18 different upper steamed distillation samples were screened to obtain SAB samples with distinct cellar aromas using the Napping-UFP method. Moreover, to investigate key differential marks between the traditional and modern SAB samples, molecular sensory science was employed. Comparative aroma extract dilution analysis (cAEDA) and multiple quantitative techniques were employed to identify and quantify odorants. Further analysis through orthogonal partial least squares discriminant analysis (OPLS-DA) revealed key discriminant marker substances. Verification using aroma recombination and omission tests pinpointed key aroma compounds. Then, partial least squares regression (PLSR) was used to screen the key cellar aroma compounds in different distillation methods, providing theoretical support and foundational data for the modernisation of SAB.



FIGURE 1 | Transition from traditional upper steaming distillation to mechanical upper steaming distillation: The left image shows the traditional method, the right image depicts the modern method.

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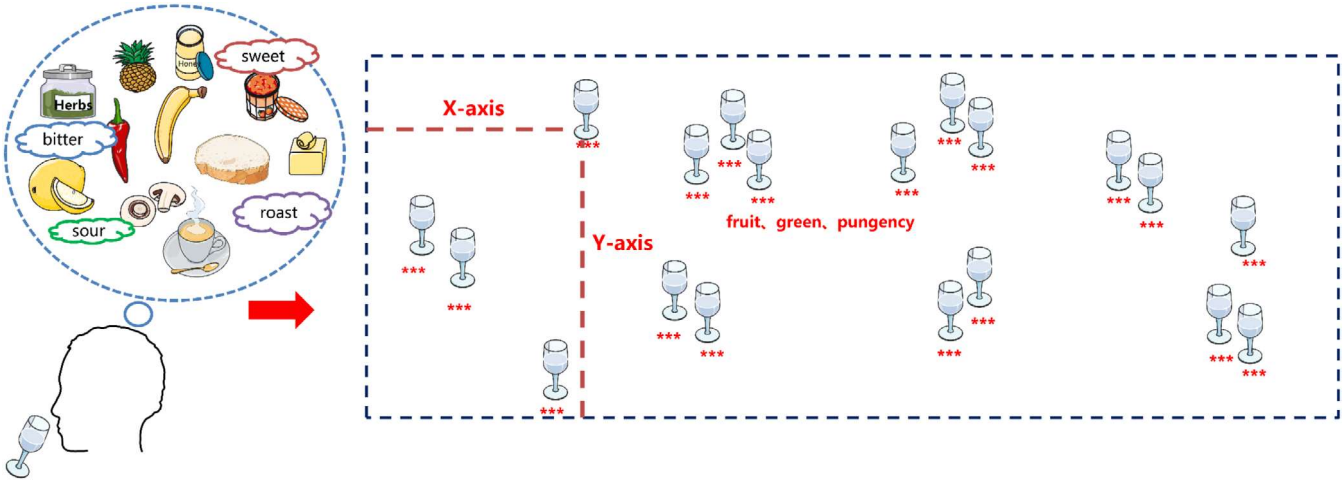


FIGURE 2 | Napping-UFP sensory experiment simulated figurative chart.

2 | Materials and Methods

2.1 | Baijiu Samples

A total of 18 base SAB samples were analysed in this study. LZLJ-T1, LZLJ-T2, LZLJ-T3, LZLJ-T4, LZLJ-T5, LZLJ-T6, LZLJ-T7, LZLJ-T8 and LZLJ-T9 are base Baijiu produced using traditional upper steaming technology. LZLJ-M1, LZLJ-M2, LZLJ-M3, LZLJ-M4, LZLJ-M5, LZLJ-M6, LZLJ-M7, LZLJ-M8 and LZLJ-M9 are base Baijiu produced using modern upper steaming technology. All samples were provided by Luzhoulaojiao Distillery Co. Ltd. (Sichuan Province, China) and stored in 250.0-mL glass bottles at 4°C until analysis.

2.2 | Chemicals and Reagents

The reference compounds (Tables 2 and 3) used for qualification and quantification had more than 95% purities (see Table S1). The specific information about the solvents is detailed in the Supporting Information.

2.3 | Sensory Evaluation

The combination of Napping with UFP was applied to obtain the holistic sensory tasks of 18 base SAB samples [32]. In our sensory experiments, ethical approval was granted by Beijing Technology and Business University Ethics Committee for Scientific Research, confirming that the study complied with institutional guidelines. All panellists provided informed consent to participate in the sensory analysis.

2.3.1 | Samples and Sample Presentation

All sensory trials tested 18 single Baijiu samples made in Sichuan Province. These Baijius were selected as the representative samples produced using modern and traditional upper steaming technologies during the second distillation stage. The alcohol content of the Baijius selected varied from 62.0% to 70.0%, with the majority being approximately 67.0%. Samples were prepared and presented following the same procedure in all sensory trials, which is detailed in the Supporting Information.

2.3.2 | Assessors

Napping-UFP was conducted following a protocol similar to that reported in previous studies [39–41], and the method for screening participants is detailed in the Supporting Information. Subsequently, 20 individuals (13 females and 7 males) were selected as sensory panellists for the formal experiment. Next, these assessors underwent training in the Napping-UFP method before conducting the formal Napping-UFP experiments.

2.3.3 | Napping-UFP Procedure

Twenty panellists were recruited to participate in the Napping exercise. The group of panellists, which met for a 2-h session, was introduced to the Napping method. They were given a 43 cm by 56 cm sheet of white paper and 18 Baijiu samples in glasses. The panellists were instructed to arrange the samples on the sheet of paper so that the similar samples were placed close together and the different samples were far apart. After Napping, panellists marked the position of each sample on the Napping paper with an **'X'** and labelled **'X'** with the sample code. For the UFP portion, panellists labelled sample positions with descriptive phrases used to characterise the sensory attributes. An example of a Napping paper with coordinates and descriptors generated by one panellist during the Napping exercise is shown in Figure 2.

2.3.4 | Descriptive Profile Analysis

The purpose of this experiment is to help sensory panellists become familiar with the aroma characteristics of each detected aroma compound in the test Baijiu samples. The analysis method is shown in the Supporting Information. After screening and discussion, sensory evaluation descriptors for LZLJ-T7 and LZLJ-M7 were finalised, including cellar, fruity, floral, sour, alcohol, sweet, grain and grassy aroma.

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Subsequently, the intensity of each aroma attribute and the overall aroma of LZLJ-T7 and LZLJ-M7 were assessed, and the flavour profile wheels of the samples were plotted. The sensory panellists initially received training in aroma intensity scaling, with reference materials and intensity standards for each aroma descriptor listed in Table 1. In the formal experiment, each sample was sniffed three times and scored from 0 to 3.

2.4 | Aroma Extraction and Separation

Based on the preliminary research, glassware cleaning, reagent pretreatment, liquid–liquid extraction (LLE) and isolation pH–LLE experiments were performed [8, 42]. The method descriptions are detailed in the Supporting Information.

2.5 | Identification Using GC–MS/GC–O–MS

The extractions above are transferred to the injection vial for GC–MS/GC–O–MS analysis. The analysis parameters are described in the Supporting Information. Three trained panellists from BTBU were selected to perform the GC–O analysis. During the GC operation, the sensory personnel recorded the aroma, intensity, and retention time while sniffing through the nose at all times. Aroma-active compounds were identified using MS detection, retention index (RI), standards and aroma characteristics. The NIST 20.0 and the Baijiu flavour compound database (built by BTBU) were used for accurate qualitative analysis. RI was calculated using a C_8–C_{30} *n*-alkane mixture under the same injection conditions as that for the Baijiu samples.

TABLE 1 | Sensory descriptions and reference samples for LZLJ-T7 and LZLJ-M7.

Aroma descriptions	Reference samples[a]	Intensity
Cellar	10 mL of the distinctive cellar aroma of SAB	5.0
Fruity	5530.0 mg/L of ethyl hexanoate	4.0
Floral	2890.0 mg/L of phenethyl alcohol	5.0
Sweet	9.1 mg/L of γ-nonalactone	5.0
Sour	The mixture of 16,000.0 mg/L acetic acid and 96.4 mg/L butyric acid	4.5
Grain	10.0 g of sorghum powder	4.0
Grassy	10.0 g of freshly crushed grass	3.0
Alcohol	65% ABV ethanol-water solution	3.0

[a]Diluent solution is ethanol.

2.6 | Comparative Aroma Extract Dilution Analysis (cAEDA)

Samples of concentrated extracts were sniffed by three trained panellists using the DB-WAX column and HP-5 columns GC–O. CH_2Cl_2 was used as the dilution solvent for aroma extracts. AEDA was performed according to a multiple of 3, and the sample was diluted until the sniffer could no longer smell any odour. Each aroma-active compound had its flavour dilution (FD) factor representing the maximum dilution factor for the smell, which could be used as one of the indicators to characterise the importance of the compound to the overall aroma of the sample.

2.7 | Multiple Quantitative Analysis of Aroma Compounds

This study employs the quantitative methods previously developed by our research group to perform quantitative analysis of aroma compounds with FD ≥ 9 in the Baijiu samples [8], detailed in Table S2.

2.7.1 | Direct Injection Combined With GC–FID Analysis

As shown in Table S2, the major compounds, including ethyl acetate, ethyl hexanoate, ethyl butyrate, acetic acid, 3-methyl-1-butanol, butyric acid and hexanoic acid at higher concentrations, were quantified using GC-FID. The details of the direct injection method and GC-FID analytical parameters are described in the Supporting Information. Compound standard solutions with various concentrations were prepared in a 65% ethanol solution. IS1 (final concentration of 229.0 mg/L) was added, and then GC–FID analysis was performed in the same manner as the Baijiu samples. All experiments were conducted three times.

2.7.2 | LLME Combined With GC–MS Analysis

According to the method published previously by our group [2], minor compounds were quantitated using LLME–GC–MS. The details of the LLME method and the GC–MS analytical parameters are described in the Supporting Information. The Baijiu samples were analysed in triplicate.

2.8 | Aroma Recombination

The aroma compounds with odour activity values (OAVs) ≥ 1 in the Baijiu samples were combined according to their detected concentrations to prepare two recombination models (LZLJ-T7 with 64% ABV and pH 4.5, and LZLJ-M7 with 66% ABV and pH 4.6) (Table 3). The recombined models and real samples of LZLJ-T7 and LZLJ-M7 were equilibrated for at least 20 min at room temperature before the descriptive analysis, as described in the 'Descriptive profile analysis' section.

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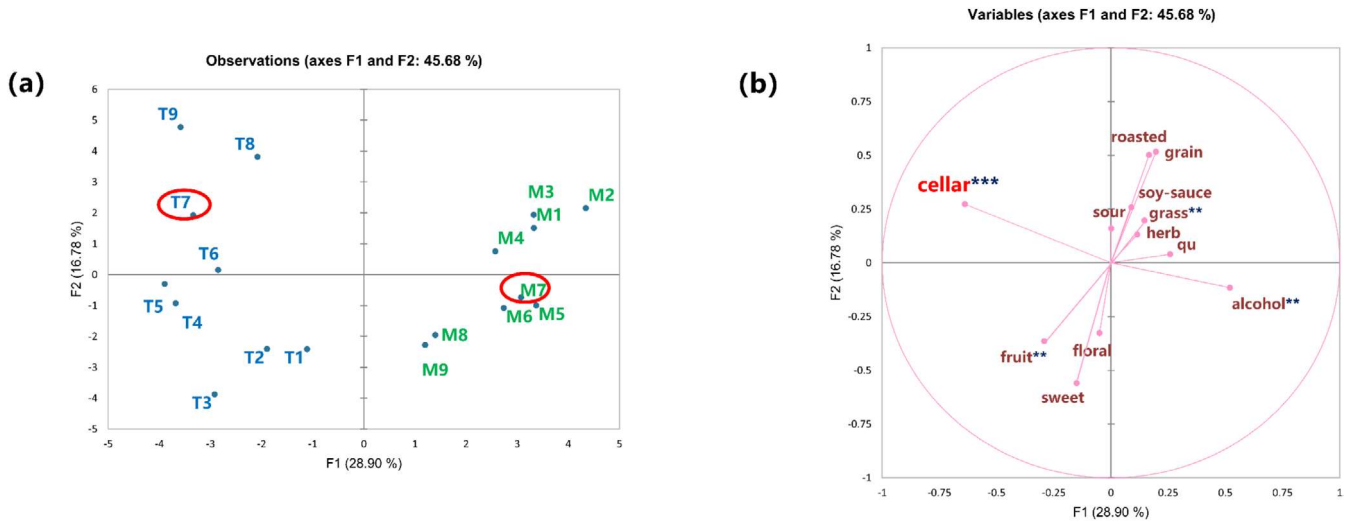


FIGURE 3 | Multiple factor analysis (MFA) of sensory parameters results (Napping + UFP) for traditional and modern technologies Baijiu samples. Sample plot (a) and attribute plot (b).

2.9 | Aroma Omission

In this experiment, 54 aroma deletion models were achieved by omitting one or more specific aroma compounds in aroma reconstitution experiments. Triangle tests were then used to distinguish the omission models from the reconstitution models, with each group of samples randomly coded with three-digit numbers. Twenty trained sensory panellists were asked to identify the solution that was different. Each test was repeated in triplicate, and the significance of the omission models was calculated according to the method described in the 'Data analysis' section of Supporting Information.

3 | Results and Discussion

3.1 | Sensory Evaluation

As shown in Figure 3a,b, analysing the observations and variable plots from MFA showed an explanatory power of 28.90% for Dim1 and 16.78% for Dim2, together accounting for 45.68% of the overall variation. Figure 3a displays that traditional technology Baijiu samples with heavy properties were positioned on the left side of dimension 1, while modern technology samples with light properties were located on the right side of dimension 1. Figure 3b shows that dimension 1 was positively loaded by alcohol and Daqu aromas and negatively by cellar and fruit aromas. Dimension 2 was positively defined by sour, roast, grain and soy sauce aromas and negatively by flower and sweet aromas. This result coincides with those of Leanie et al. [43], who analysed the properties of liquor products using Napping and showed that Dim1 allowed a clear distinction between samples with light properties, such as 'citrus aroma' or 'fruity aroma' and those with heavy properties such as 'mud aroma', 'grass aroma', 'dark gold color', or 'dark brown color'.

The typical flavour characteristics of SAB include the predominant cellar aroma, accompanied by a comfortable blend of fragrances according to GB/T 10781.1–2021 [44]. Among Baijiu samples produced using modern technologies, those coded as M1, M2, M3 and M4 were associated with roast, grain and grass aroma terms, while those coded as M5, M6 and M7 were related to a stronger alcohol aroma. Among the Baijiu samples undergoing traditional technology, T1, T2 and T3 were characterised by fruit and sweet aromas, and those coded as T6, T7, T8 and T9 samples displayed a richer cellar aroma, as shown in Figure 3b. The higher overall aroma perception may be caused by the higher volatile compound generation in these samples. Moreover, the traditional brewing method would have contributed to the overall aroma and linking because a higher content of volatile compounds has been reported in samples submitted to the traditional brewing procedure compared with those subjected to modern technology [18]. However, different upper steaming technologies affecting the formation of aroma compounds require further in-depth research.

Moreover, Figure 3b indicates that the cellar aroma branch has the maximum projection length and the smallest angle with the X-axis in one dimension. This suggests its highest contribution to one dimension and confirms significant flavour differences between traditional and modern technology Baijiu samples. Moreover, the flavour trend of T7 is consistent with the cellar aroma branch, exhibiting rich cellar aroma, while M7 lacks prominence in cellar aroma. Based on sensory data analysis, T7 and M7 samples were selected as target samples for the subsequent experiments.

3.2 | Aroma-Active Compounds in LZLJ-T7 and LZLJ-M7

cAEDA on the NBF and AF from each tested Baijiu sample obtained using LLE were analysed to explain the sources of substances causing flavour differences between LZLJ-T7 and LZLJ-M7 and to explore their roles in flavour formation. As shown in Table 2, 74 and 65 aroma regions were detected in LZLJ-T7 and LZLJ-M7, respectively, with FD values in the range of1 to 59,0499. According to the qualitative methods described in section 2.5, 73 and 64 aroma-active compounds were accurately identified in LZLJ-T7 and LZLJ-M7. Through comparative analysis, 57 common aroma compounds were found.

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TABLE 2 | Aroma-active compounds in LZLJ-T and LZLJ-M7.

NO.*	Aroma compounds[a]	Ordor description	DB-WAX		HP-5MS		FD factors[c]		Fraction[d]
			RI	LRI[b]	RI	LRI[b]	LZLJ-M7	LZLJ-T7	
Esters									
1	Ethyl acetate	Aromatic, Brandy, Grape	893	887	600	605	2187	243	NBF
2	Ethyl butyrate	Apple, Strawberry	1030	1041	807	803	19,683	59,049	NBF + AF
3	Ethyl Hexanoate	Fruit, Pineapple	1233	1239	1006	1002	59,049	59,049	NBF + AF
4	Isoamyl acetate	Apple, Banana	1110	1126	855	876	243	243	NBF + AF
5	Methyl hexanoate	Ester, Fresh	1150	1181	929	925	3	27	NBF
6	Hexyl acetate	Apple Peel, Peach	1253	1276	1021	1014	81	9	NBF
7	Caproic acid propyl ester	Pineapple, Roasted Garlic	1300	1324	1099	1089	27	27	NBF
8	Butyl hexanoate	Fruit, Grass, Green	1393	1418	1195	1188	27	9	NBF
9	Pentyl hexanoate	Floral, Fresh, Rose	1504	1525	1291	1287	81	81	NBF
10	Ethyl nonanoate	Floral	1477	1526	1299	1294	3	9	AF
11	Ethyl benzoate	Flower, Fruit	1667	1652	1171	1171	27	9	NBF
12	Phenethyl acetate	Flower, Honey, Rose	1828	1822	1258	1256	27	9	NBF
13	FEMA 2575	Green, Herb, Oil	1851	1796	1587	1582	243	3	NBF
14	Butyl acetate	Apple, Banana	1080	1059	833	815	3	243	NBF + AF
15	Ethyl valerate	Apple, Fruit	1145	1139	908	898	59,049	19,683	NBF + AF
16	Ethyl heptanoate	Fruit, Grape	1355	1352	1103	1093	3	243	NBF
17	Ethyl L(−)-lactate	Cheese, Floral, Fruit	1301	1356	847	/	3	27	NBF
18	Ethyl caprylate	Pear, Pineapple	1429	1435	1202	1195	243	19,683	NBF
19	2-hydroxy-4-methylvalerate	Green, fruity, apple	1534	1538	1063	1060	81	243	NBF
20	Diethyl succinate	Floral, Fruit, Wine	1674	1687	1186	1182	27	3	NBF
21	Ethyl phenylacetate	Floral, Honey, Rose	1795	1785	1246	1247	9	3	NBF
22	Ethyl 3-phenylpropionate	Flower, Honey	1908	1900	1349	1350	81	243	NBF + AF
23	FEMA 2721	Fruit	1110	1136	855	861	3	3	NBF
24	Ethyl laurate	Floral, Fruit, Leaf	1868	1856	1597	1597	3	3	NBF + AF
25	Ethyl myristate	Wax	2038	2012	1797	1796	3	3	NBF
26	Ethyl isovalerate	Pineapple, Sour	1056	1072	885	859	3	/	NBF + AF
27	Isoamyl hexanoate	Anise, Fruit, Spice	1446	1469	1253	1254	3	/	NBF
28	Hexyl hexanoate	Apple Peel, Peach	1612	1619	1389	1385	3	3	NBF + AF
29	Ethyl undecanoate	Coconut, Cognac	1753	1732	1498	1459	3	/	NBF
30	Ethyl (Z)hex-3-enoate	Fruit	1310	1295	990	993	/	3	NBF
31	Butyl valerate	Fruity	1330	1336	1056	1063	/	3	NBF
32	Heptyl acetate	Floral, Fresh	1363	1364	1124	1112	/	3	NBF + AF
33	Cyclohexanecarboxylic acid ethyl ester	Fruit	1408	1413	1193	1130	/	3	NBF + AF
34	Isobutyl hexanoate	Green, Spice	1355	1347	1154	1148	/	3	NBF

(Continues)

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TABLE 2 | (Continued)

NO.*	Aroma compounds[a]	Ordor description	DB-WAX		HP-5MS		FD factors[c]		Fraction[d]
			RI	LRI[b]	RI	LRI[b]	LZLJ-M7	LZLJ-T7	
35	Ethyl trans-4-decenoate	Fruit	1654	1680	1382	1311	/	3	NBF
36	FEMA 3222	Fruit	2350	2373	1900	1854	/	3	NBF
Alcohols									
37	3-Methyl-1-butanol	Burnt, Floral, Malt	1208	1211	773	730	729	2187	NBF + AF
38	1-Octen-3-ol	Cucumber, Mushroom	1426	1456	991	986	3	27	NBF
39	1-Butanol	Fusel oil, sweet	1145	1150	703	668	27	729	NBF + AF
40	1-Hexanol	Flower, Grass, Herb	1348	1359	886	871	2187	27	NBF
41	Furfuryl alcohol	Burnt, Caramel	1650	1665	890	864	81	81	NBF
42	Phenethyl alcohol	Fruit, Honey, Rose	1955	1923	1115	1114	729	729	NBF + AF
43	2-Methyl-1-propanol	ethereal winey cortex	1110	1094	598	622	27	27	NBF
44	2-Pentanol	Fusel Oil, Green	1122	1119	710	706	3	3	NBF + AF
45	1-Octanol	Bitter Almond, Fat, Floral	1551	1564	1050	1068	3	/	NBF
46	2-Butanol	Fruit	1029	1030	605	610	/	3	AF
47	1-Propanol	Alcohol, Candy	1030	1035	525	532	/	3	NBF + AF
48	2-Ethylhexanol	Green, Rose	1473	1484	1024	1031	/	3	NBF
49	1-Nonanol	Fat, Floral, Green, Oil	1670	1666	1207	1186	/	3	NBF
Acids									
50	Butyric Acid	Sharp, acetic, cheese	1609	1604	875	848	729	729	AF
51	Hexanoic acid	Cheese, Pungent, Sour	1860	1861	943	981	9	27	NBF + AF
52	Acetic acid	Acid, Vinegar	1398	1461	654	660	27	243	AF
53	Isovaleric acid	Cheese, Pungent	1663	1647	907	875	9	27	AF
54	Valeric acid	Acidic, sweaty, rancid	1735	1755	944	906	9	27	AF
55	Heptanoic acid	Apricot, Floral, Sour	1975	1953	1094	1080	27	3	AF
56	Isobutyric acid	Burnt, Butter, Cheese	1553	1564	763	785	3	3	AF
57	Propionic acid	Fat, Pungent, Soy	1517	1535	727	740	3	/	AF
58	Nonanoic acid	Fat, Green, Sour	2220	2169	1301	1297	3	/	AF
59	Phenylacetic acid	Caramel, Flower, Honey	2544	2521	1278	1254	/	3	AF
Carbonyls									
60	(E)-2-Octenal	Grass, Green, Spice	1411	1434	1042	1057	9	3	NBF
61	Phenylacetaldehyde	Honey, Nut, Pungent	1636	1646	1045	1044	81	81	NBF
62	Trans-2-dodecenal	Fruit	1864	1830		1464	3	3	NBF
63	1-Nonanal	Floral, Green, Lemon	1377	1396	1107	1102	3	3	NBF
64	trans, trans-2,4-Decadien-1-al	Fried, Fat, Oxidised	1825	1819	1332	1318	3	3	NBF
65	(Z)-non-2-enal	Paper	1497	1543	1127	1147	3	/	NBF
66	2-Nonanone	Weedy, earthy, herbal	1376	1393	1095	1091	/	3	NBF

(Continues)

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TABLE 2 | (Continued)

NO.*	Aroma compounds[a]	Ordor description	DB-WAX		HP-5MS		FD factors[c]		Fraction[d]
			RI	LRI[b]	RI	LRI[b]	LZLJ-M7	LZLJ-T7	
67	Geranylacetone	Fruit	1890	1862	1459	1452	/	3	NBF
Sulphur compounds									
68	Dimethyl trisulfide	Cabbage, onion	1373	1368	1004	972	243	243	NBF
Nitrogenous compounds									
69	Tetramethylpyrazine	Cocoa, Coffee, Roast	1468	1460	1091	1087	81	27	NBF
Phenols									
70	2-Methoxy-4-methylphenol	Phenol	1976	1956	1221	1199	9	9	NBF + AF
71	4-Ethyl-2-methoxyphenol	Clove, Phenol, Spice	2057	2032	1315	1280	27	81	NBF + AF
72	p-Cresol	Phenolic, animal mimosa	2111	2091	1079	1086	9	9	NBF + AF
73	Vanillin	Vanilla	2534	2566	1360	1394	9	3	NBF + AF
74	4-Ethyl-2-methoxyphenol	Clove, Phenol, Spice	2057	2032	1296	1280	3	3	NBF + AF
Lactones									
75	gamma-Nonanolactone	Coconut, sweet, buttery	2008	2012	1362	1362	3	9	NBF + AF
76	4,5-Dimethyl-3-hydroxy-2,5-dihydrofuran-2-one	Cotton Candy, Spice	2200	2204	1145	1105	3	3	NBF + AF
77	Gamma Butyrolactone	Caramel, Roasted Nut	1627	1643	900	915	/	3	NBF + AF
78	Undecanolactone	Peach	2321	2345	1550	1539	/	3	NBF
Furans									
79	2-(ethoxymethyl)furan	Nuts	1287	1297	/	/	243	2187	NBF + AF
Others									
80	1,1-Diethoxy-3-methylbutane	Fruit	1033	1062	958	955	3	1	NBF + AF
81	Unknown	Fusel Oil	/	/	/	/	19,683	2187	NBF + AF

[a]Aroma compounds: Aroma-active compounds are identified by comparing their retention indices on DB-WAX and HP-5MS columns, aroma characteristics, and mass spectrometry information with those of the standards, as well as against the NIST 20.0 library.
[b]LRI: RI in literature.
[c]FD factors: Dilution factors determined by AEDA on the DB-WAX column.
[d]Fraction: NBF, neutral/basic fraction; AF, acidic fraction.

Among these compounds, ethyl hexanoate (fruity, pineapple), ethyl butyrate (apple, strawberry), ethyl valerate (apple, fruity) and ethyl octanoate (pineapple, pear) have the highest FD values, ranging from 19,893 to 59,049, in both LZLJ-T7 and LZLJ-M7. These compounds are characteristic aroma compounds in SAB, primarily contributing to the fruity and sweet aromas. In addition, ethyl acetate, isoamyl acetate, hexyl octanoate, ethyl butyrate, ethyl heptanoate, DL-2-hydroxy-4-methylpentanoate, ethyl 3-phenylpropionate, isoamyl alcohol, *n*-butanol, phenylethyl alcohol, *n*-hexanol, butyric acid, acetic acid, dimethyl trisulfide and ethyl furfuryl ether exhibit relatively high FD values ($243 \leq FD < 2187$) in both samples. It is inferred that these compounds are important aroma components in SAB distilled using traditional and modern technologies, significantly contributing to the ester, floral, grassy, malty, nutty, roasted, onion and acidic flavours. The detailed comparison of FD values of the aroma profiles in two Baijiu samples is described in the Supporting Information.

3.3 | Quantitative Analysis and OAVs

3.3.1 | Odorant Concentrations

As shown in Table 3, ethyl acetate, ethyl hexanoate, ethyl butyrate, isoamyl alcohol, butyric acid and hexanoic acid are the core components of LZLJ-T7 and LZLJ-M7. These concentrations of the components were all present at levels above 100 mg/L. Among them, ethyl acetate has the highest concentration (2926.0–4021.9 mg/L). Besides, ethyl pentanoate (16.1–18.0 mg/L), pentanoic acid (49.4–57.1 mg/L), butanol

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TABLE 3 | Concentrations and OAVs of LZLJ-T7 and LZLJ-M7.

NO.	Compounds	Odour threshold (μg/L)	Concentrations (μg/L) ± SD ($n = 3$) LZLJ-M7	LZLJ-T7	OAV LZLJ-M7	LZLJ-T7
1	Ethyl Hexanoate	55.33[a]	873,258 ± 9159	738,302 ± 3649	15,783	13,344
2	Ethyl butyrate	81.50[a]	165,929 ± 4181	181,049 ± 3783	2036	2221
3	Ethyl valerate	26.80[a]	18,031 ± 270	16,080 ± 161	673	600
4	Dimethyl trisulfide	0.40[a]	139 ± 0.4	136 ± 0.1	347	339
5	1-Octen-3-ol	0.00612[a]	1 ± 0.2	1 ± 0.4	189	194
6	(E)-2-Octenal	0.0151[a]	2 ± 0.3	2 ± 0.1	140	136
7	Butyric Acid	964.00[a]	128,673 ± 749	81,517 ± 939	133	85
8	Valeric acid	389.00[a]	49,388 ± 4109	57,055 ± 762	127	147
9	Ethyl acetate	32600.00[a]	4,021,850 ± 43,275	2,926,064 ± 56,016	123	90
10	2-(ethoxymethyl)furan	10.71[a]	940 ± 2	1121 ± 4	88	105
11	ethyl 2-hydroxy-4-methylvalerate	55.30[a]	3374 ± 13	4326 ± 42	61	78
12	Ethyl caprylate	12.90[a]	940 ± 7	1672 ± 79	73	130
13	Hexanoic acid	2517.16[a]	82,362 ± 716	123,135 ± 7632	33	49
14	Isoamyl acetate	93.93[a]	2885 ± 81	2236 ± 78	31	24
15	1-Butanol	2730[a]	54,922 ± 574	67,567 ± 680	20	25
16	Methyl hexanoate	41.00[a]	336 ± 7	357 ± 7	8	9
17	Phenylacetaldehyde	262.00[a]	1733 ± 19	1783 ± 0.6	7	7
18	4-Ethyl-2-methoxyphenol	123.00[a]	524 ± 0.5	526 ± 0.3	4	4
19	Isovaleric acid	1045.47[a]	3123 ± 383	4507 ± 246	3	4
20	1-Hexanol	5370.00[a]	20,155 ± 59	15,678 ± 26	4	3
21	3-Methyl-1-butanol	179190.83[a]	379,229 ± 1775	316,720 ± 4522	2	2
22	2-Methoxy-4-methylphenol	315.00[a]	579 ± 0.2	579 ± 0.4	2	2
23	gamma-Nonanolactone	91.00[a]	130 ± 0.8	13 ± 1	1	1
24	Furfuryl alcohol	2000.00[a]	2533 ± 7	2505 ± 2	1	1
25	Ethyl phenylacetate	407.00[a]	515 ± 5	457 ± 2	1	1
26	Ethyl 3-phenylpropionate	125.00[a]	148 ± 0.1	213 ± 4	1	2
27	Butyl hexanoate	678.00[a]	675 ± 16	848 ± 30	1	1
28	Butyl acetate	4600.00[a]	4971 ± 42	4316 ± 27	1	1
29	Phenethyl acetate	909.00[a]	510 ± 0.4	493 ± 0.7	< 1	< 1
30	Hexyl acetate	1500.00[a]	762 ± 4	756 ± 9	< 1	< 1
31	Vanillin	439.00[a]	265 ± 8	245 ± 31	< 1	< 1
32	p-Cresol	167.00[a]	95 ± 0.7	103 ± 5	< 1	< 1
33	Acetic acid	160000.00[a]	35,645 ± 2856	41,356 ± 2385	< 1	< 1
34	Ethyl benzoate	1430.00[a]	451 ± 3	442 ± 1.2	< 1	< 1
35	Ethyl heptanoate	13200.00[a]	2031 ± 30	2398 ± 19	< 1	< 1

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TABLE 3 | (Continued)

NO.	Compounds	Odour threshold (μg/L)	Concentrations (μg/L) ± SD ($n = 3$) LZLJ-M7	LZLJ-T7	OAV LZLJ-M7	LZLJ-T7
36	Heptanoic acid	13300.00[a]	1937 ± 205	1731 ± 165	< 1	< 1
37	Ethyl nonanoate	3150.61	429 ± 1.3	446 ± 0.7	< 1	< 1
38	Ethyl L(−)-lactate	128000.00	7565 ± 81	8505 ± 167	< 1	< 1
39	Ethyl L(−)-lactate	12800.00[a]	548 ± 7	634 ± 16	< 1	< 1
40	Phenethyl alcohol	28900.00[a]	994 ± 0.3	989 ± 7	< 1	< 1
41	Tetramethylpyrazine	80073.16[a]	449 ± 1	442 ± 4	< 1	< 1
42	Diethyl succinate	353193.00[a]	1380 ± 10	1087 ± 33	< 1	< 1
43	Pentyl hexanoate	/[b]	433 ± 0.8	448 ± 3	/	/
44	Fema 2575	/[b]	492 ± 0.3	500 ± 9	/	/

[a]Olfactory threshold in the 46% ethanol-water solution previously reported in Ref.(8).
[b]/, not detected.

(54.9–67.6 mg/L), hexanol (15.7–20.2 mg/L) and acetic acid (35.6–41.4 mg/L) are also present at concentrations exceeding 10 mg/L. Moreover, DL-2-hydroxy-4-methylpentanoate (3.4–4.3 mg/L), ethyl octanoate (0.9–1.7 mg/L), isoamyl acetate (2.2–2.9 mg/L), benzaldehyde (1.7–1.8 mg/L), isovaleric acid (3.1–4.5 mg/L), furfuryl alcohol (2.5 mg/L), ethyl butyrate (4.3–5.0 mg/L), heptanoic acid (1.7–1.9 mg/L), ethyl L(−)-lactate (7.6–8.5 mg/L) and diethyl succinate (1.1–1.4 mg/L) are also important aroma compounds. These compounds are present at relatively high concentrations in LZLJ-T7 and LZLJ-M7 analysed in this study.

3.3.2 | OAVs

This section further examines the aroma expression of aroma compounds in LZLJ-T7 and LZLJ-M7 using OAVs, with 28 aroma compounds with OAV ≥ 1, shown in Table 3. In LZLJ-M7, ethyl hexanoate has the highest OAV value of 15,783. Other compounds with high OAV values (OAV ≥ 100) include ethyl butyrate (OAV ≥ 2036), ethyl valerate (OAV ≥ 673), dimethyl trisulfide (OAV ≥ 467), 1-octen-3-ol (OAV ≥ 189), *trans*-2-octenal (OAV ≥ 136), butyric acid (OAV ≥ 133), valeric acid (OAV ≥ 127) and ethyl acetate (OAV ≥ 123), indicating their significant contribution to the flavour characteristics of LZLJ-M7. In LZLJ-T7, ethyl hexanoate has the highest OAV value (OAV ≥ 13,344). Ethyl butyrate (OAV ≥ 2221), ethyl valerate (OAV ≥ 600), dimethyl trisulfide (OAV ≥ 339), 1-octen-3-ol (OAV ≥ 194), *trans*-2-octenal (OAV ≥ 140), valeric acid (OAV ≥ 147), ethyl octanoate (OAV ≥ 130), and ethyl furfuryl ether (OAV ≥ 105) also have OAV values greater than 100, making them important flavour compounds in LZLJ-T7. Finally, the comparison of important flavour compounds in LZLJ-T7 and LZLJ-M7 is detailed in the Supporting Information.

3.3.3 | OPLS-DA Analysis

The OPLS-DA analysis was conducted on the aroma-active compounds (OAV ≥ 1) in LZLJ-T7 and LZLJ-M7, as shown in Figure 4. The OPLS-DA model has an R^2X of 0.866, R^2Y of 0.998, and Q^2 of 0.989 > 0.5, indicating a good fit. The two groups of samples can be completely distinguished in Figure 4a. Among the main aroma compounds in LZLJ-T7 and LZLJ-M7, 18 aroma-active components with variable importance in the projection (VIP) > 1 were selected (Figure 4b), including ethyl acetate, ethyl hexanoate, isoamyl acetate, butyl hexanoate, ethyl butyrate, ethyl valerate, ethyl octanoate, DL-2-hydroxy-4-methylpentanoate, ethyl phenylacetate, ethyl 3-phenylpropionate, isoamyl alcohol, *n*-butanol, *n*-hexanol, furfuryl alcohol, isovaleric acid, dimethyl trisulfide, 4-ethylguaiacol and ethyl furfuryl ether. These compounds can be used as markers for distinguishing SABs produced using modern and traditional upper steaming technologies.

3.4 | Aroma Recombination and Omission Experiments

3.4.1 | Reconstruction of the Aroma Profiles of LZLJ-T7 and LZLJ-M7

To verify the flavour contribution of aroma compounds to LZLJ-T7 and LZLJ-M7, this study added 28 main aroma compounds with OAV ≥ 1 at the concentrations listed in Table 3 to prepare the reconstituted samples. Descriptive profile tests were performed separately for each aroma reconstruction compared with their corresponding original samples by rating the intensity of the eight aroma attributes and the overall similarity (Figure 5). The overall aroma similarity between the reconstituted models and the original samples of LZLJ-T7 and LZLJ-M7 was relatively high, reaching 2.7 and 2.5 points (out of 3.0), respectively. As for LZLJ-M7, the reconstituted model and the original sample showed the most similar scores in fruit, grass, and alcohol aroma attributes, and cellar, sweet, and floral aromas were similar to those of LZLJ-T7. The reconstituted model samples accurately reflected the flavour profiles of LZLJ-T7 and LZLJ-M7 in terms of cellar, fruit, floral, sour, alcohol, sweet, grain and grassy aroma attributes.

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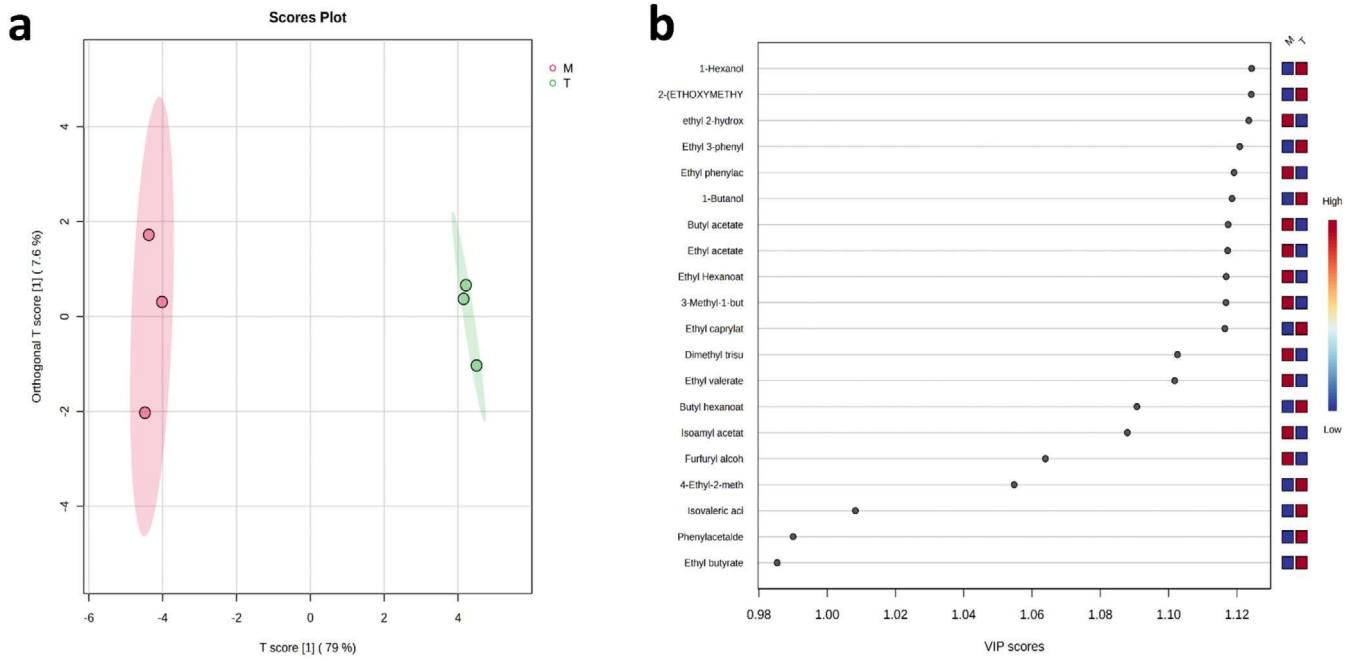


FIGURE 4 | OPLS-DA analysis of main aroma compounds in LZLJ-T7 and LZLJ-M7. score plot (a), VIP plot (b).

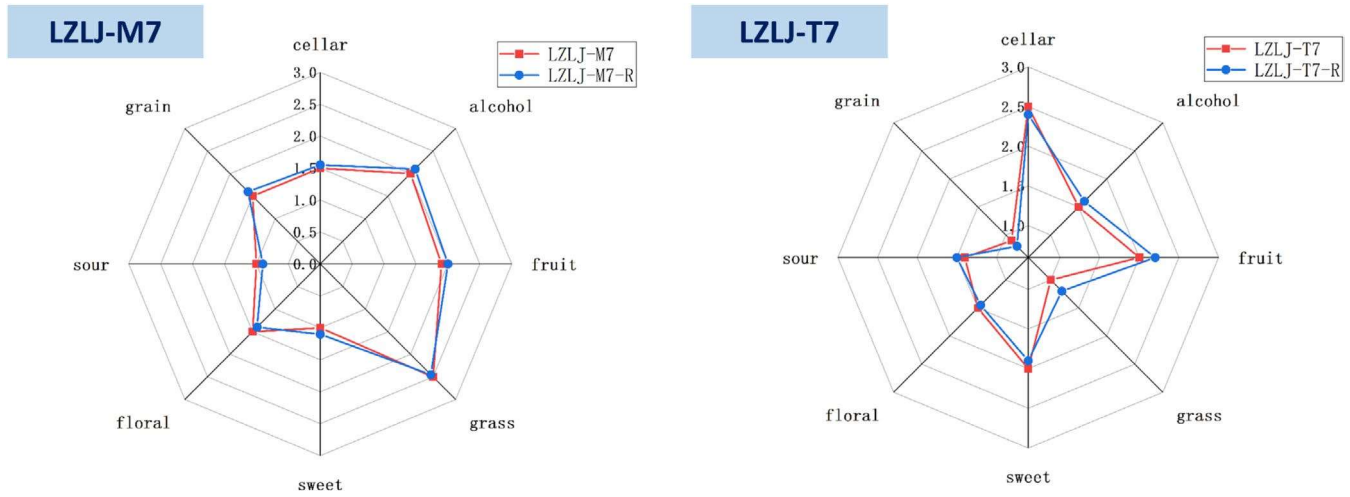


FIGURE 5 | Aroma profiles and reconstitution models of LZLJ-T7 and LZLJ-M7.

3.4.2 | Omission Studies

To investigate the contribution of certain aroma compounds to the overall cellar aroma profile of LZLJ-T7 and LZLJ-M7, 54 omission models were prepared in which either a single component or a group of components with similar chemical structures and aroma descriptions were omitted (Table 4). Each omission model was evaluated using the triangle test compared with the complete model. The impact of omitting specific aromatic compounds in the omission models (MH1 to MH8) is described in the Supporting Information.

The results of the omission experiments indicate that ethyl hexanoate, ethyl acetate, ethyl pentanoate, ethyl octanoate, ethyl phenylacetate, isoamyl acetate, DL-2-hydroxy-4-methylpentanoate, methyl hexanoate, ethyl butyrate, ethyl 3-phenylpropanoate, ethyl hexanoate, γ-nonalactone, hexanol and dimethyl trisulfide are the key aroma compounds for LZLJ-M7. These compounds significantly contribute to the overall fruity, grassy and alcoholic aromas. Ethyl hexanoate, DL-2-hydroxy-4-methylpentanoate, ethyl acetate, ethyl octanoate, ethyl 3-phenylpropanoate, ethyl hexanoate, ethyl butyrate, ethyl phenylacetate, methyl hexanoate, hexanol, isoamyl alcohol, pentanoic acid, isovaleric acid, butyric acid, hexanoic acid, dimethyl trisulfide and 2-phenylacetaldehyde are identified as the key aroma compounds for LZLJ-T7. These compounds significantly contribute to the overall cellar, fruity, sweet and acidic aged aromas (Figure 6).

3.4.3 | Analysis of Key Differentiating Compounds and Aroma Attributes

This section integrates the key differential substances with VIP > 1 screened through OPLS-DA and the key aroma compounds

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TABLE 4 | Omission experiments from LZLJ-T7 and LZLJ-M7 complete models.

No.	Omission Compounds	Right of selection[a]	LZLJ-M7[b]	Right of selection[a]	LZLJ-T7[b]
M_H1	All esters	17/20	***	16/20	***
M_H1-1	Ethyl acetate	13/20	**	14/20	***
M_H1-2	Isoamyl acetate	13/20	**	4/20	\
M_H1-3	Ethyl butyrate	10/20	\	13/20	**
M_H1-4	Ethyl Hexanoate	15/20	***	17/20	***
M_H1-5	2-hydroxy4-methyl ethyl valerate	13/20	**	11/20	*
M_H1-6	Ethyl caprylate	13/20	**	11/20	*
M_H1-7	Ethyl valerate	13/20	**	9/20	\
M_H1-8	Ethyl phenylacetate	12/20	*	15/20	***
M_H1-9	Methyl hexanoate	11/20	*	11/20	*
M_H1-10	Butyl acetate	12/20	*	9/20	\
M_H1-11	Ethyl 3-phenylpropionate	11/20	*	11/20	*
M_H1-12	Butyl hexanoate	11/20	*	12/20	*
M_H2	All alcohols	11/20	*	14/20	***
M_H2-1	1-Butanol	7/20	\	9/20	\
M_H2-2	3-Methyl-1-butanol	8/20	\	12/20	*
M_H2-3	Furfuryl alcohol	9/20	\	8/20	\
M_H2-4	1-Hexanol	12/20	*	11/20	*
M_H2-5	1-Octen-3-ol	8/20	\	7/20	\
M_H3	All acids	9/20	\	12/20	*
M_H3-1	Butyric Acid			13/20	**
M_H3-2	Hexanoic acid			12/20	*
M_H3-3	Valeric acid			13/20	**
M_H3-4	Isovaleric acid			13/20	**
M_H4	Dimethyl trisulfide	11/20	*	12/20	*
M_H5	All carbonyls	7/20	\	12/20	*
M_H5-1	Phenylacetaldehyde			13/20	**
M_H5-2	(E)-2-Octenal			5/20	\
M_H6	All phenols	7/20	\	8/20	\
M_H6-1	4-Ethyl-2-methoxyphenol				
M_H6-2	2-Methoxy-4-methylphenol				
M_H7	gamma-Nonanolactone	14/20	***	10/20	\
M_H8	ethyl furfuryl ether	8/20	\	8/20	\

[a]Number of correct judgements in omission experiments from 20 assessors by the triangle test.
[b]Significance: *, significant ($p \leq 0.05$); **, highly significant ($p \leq 0.01$); ***, very highly significant ($p \leq 0.001$).

of LZLJ-T7 and LZLJ-M7 to obtain key differential flavour compounds (Figure S2): ethyl hexanoate, ethyl acetate, ethyl octanoate, ethyl phenylacetate, DL-2-hydroxy-4-methylpentanoate, ethyl 3-phenylpropanoate, ethyl hexanoate, ethyl pentanoate, isoamyl acetate, ethyl butanoate, hexanol, isoamyl alcohol, isovaleric acid and dimethyl trisulfide.

To further analyse the relationship between the key differential aroma compounds of LZLJ-T7 and LZLJ-M7 and aroma attributes, PLSR was used. OAVs of the aroma-active compounds were used as Y variables, and the scores of aroma attributes were used as X values to analyse the relationship between Y and X. As shown in Figure 7, DL-2-hydroxy-4-methylpentanoate and

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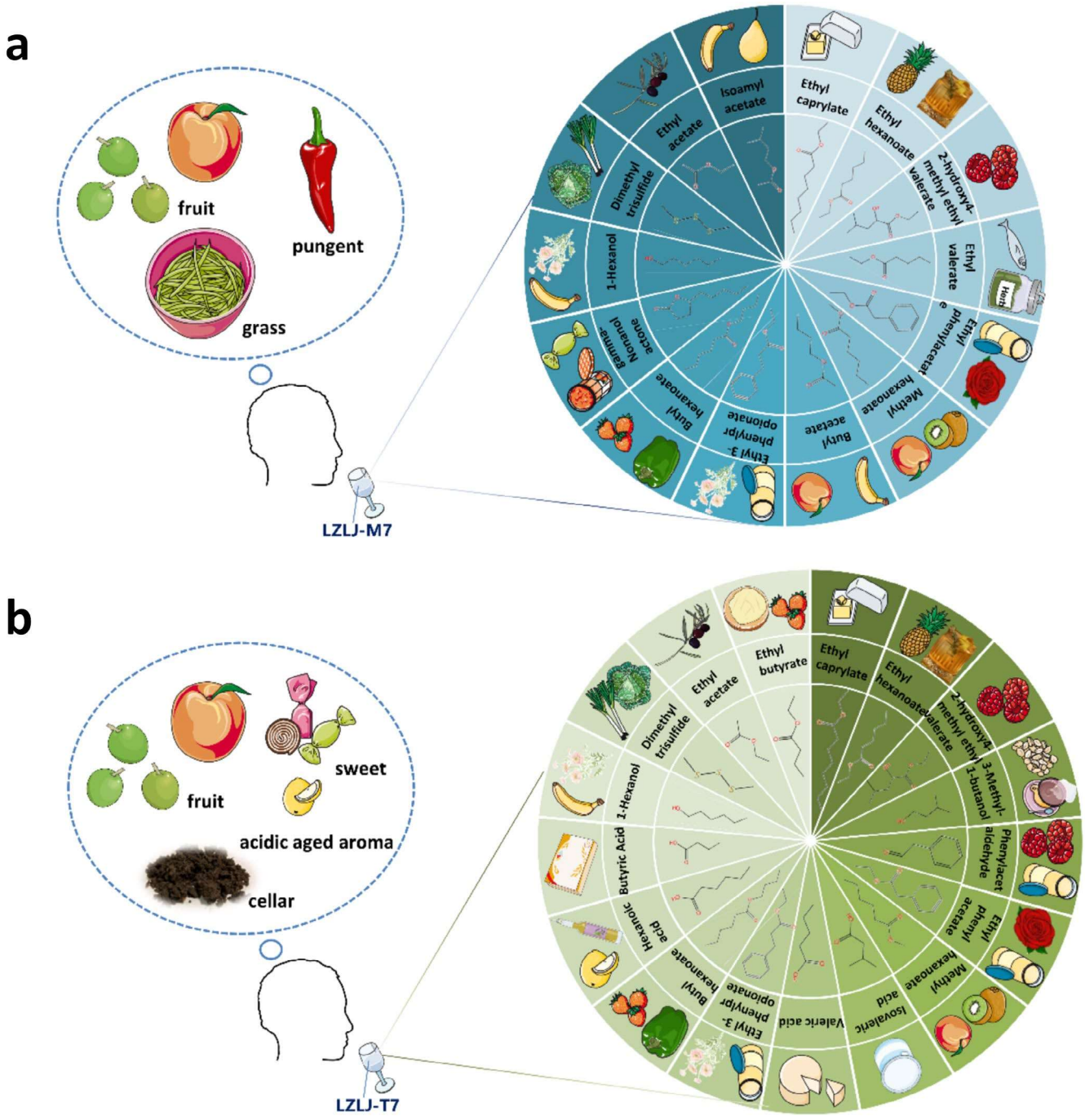


FIGURE 6 | Flavour profile of key aroma compounds in LZLJ-M7 (a) and LZLJ-T7 (b).

ethyl phenylacetate are located in the second quadrant, contributing significantly to floral aroma. Hexanol is located in the third quadrant, most related to alcohol aroma. Ethyl acetate, isoamyl alcohol, ethyl pentanoate, isoamyl acetate, ethyl hexanoate and ethyl butanoate are located on the right side of the Y-axis, related to grassy, fruity and grain aromas. Among them, ethyl acetate and isoamyl alcohol contribute the most to a grassy aroma, as isoamyl alcohol also exhibits fruity and grassy aromas in fruit wine [45]. Ethyl pentanoate, isoamyl acetate and ethyl hexanoate are most related to a fruity aroma, similar to reports in light aroma type Baijiu [46], while ethyl butanoate contributes the most to grain aroma. Ethyl hexanoate, ethyl 3-phenylpropanoate, isovaleric acid, dimethyl trisulfide and ethyl octanoate are located on the left side of the Y-axis, related to sweet, cellar and sour aromas. Ethyl hexanoate is highly related to cellar and sweet aromas, consistent with research on cellar aromas by Xiao et al. [7] Ethyl 3-phenylpropanoate is most related to sweet and cellar aromas, also presenting sweet aroma in Qingke Baijiu [47]. Isovaleric acid is most related to sour and cellar aromas, while dimethyl trisulfide and ethyl octanoate also have certain contributions. In conclusion, it is evident that ethyl hexanoate, ethyl 3-phenylpropanoate, isovaleric acid, dimethyl trisulfide and ethyl octanoate contribute the most to the cellar aroma among the key differential aroma compounds in base

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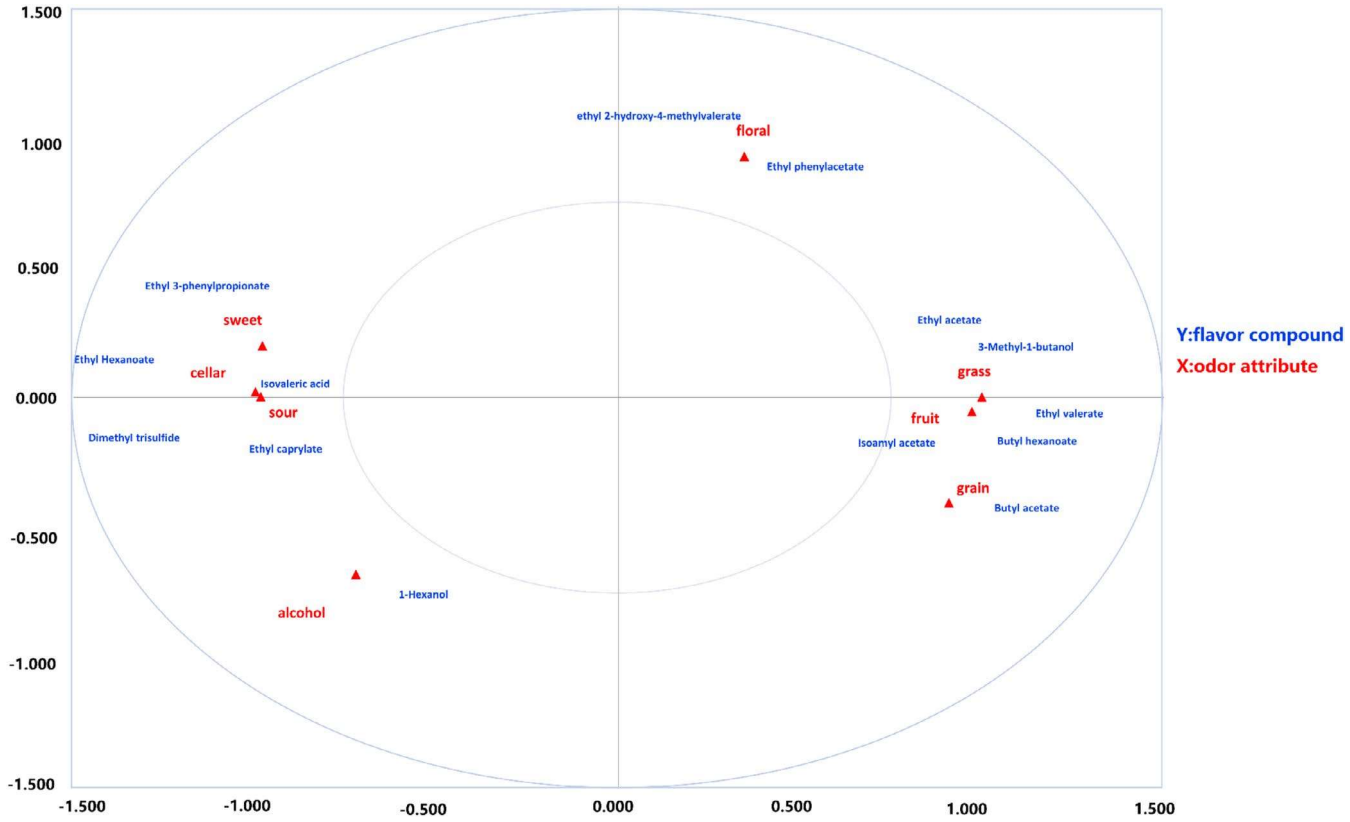


FIGURE 7 | Analysis of key differentiating compounds and aroma attributes.

SABs produced using modern and traditional upper steaming technologies.

The significant differences in the key cellar aroma compounds between traditional and modern technology samples can be largely attributed to the distinct upper steaming processes employed. Traditional upper steaming typically involves lower steam pressure and a slower, more gradual temperature ramp. Conversely, the modern upper steaming process, characterised by higher steam pressure and rapid temperature escalation, prioritises efficiency and ethanol yield [16]. This environment favours the extraction of highly volatile compounds and ethanol but may selectively limit the extraction or cause thermal degradation of heavier molecules essential for the profound, lingering cellar character [19]. The resultant aroma profile tends towards lighter, fruitier and more overtly alcoholic odours, with the complex base odours derived from the microbial ecosystem ('jiao ni') being less pronounced, as seen in the lower contribution of the cellar aroma branch and compounds like dimethyl trisulfide in the modern sample (LZLJ-M7) in Figure 3b. This mechanistic understanding, linking the physical conditions of the upper steaming process to the observed differences in volatile compound composition and sensory characteristics, is crucial for optimising process parameters to achieve desired aroma profiles in SAB production.

Although the relationship between aroma attributes and compounds in Baijiu has been extensively studied, further comprehensive considerations are needed owing to the complexity of samples and the limitations of research methods. The formation of the cellar aroma in SAB results from various factors: multiple compounds produced during fermentation and distillation, as well as the chemical reactions and interactions between compounds during ageing and storage. Its diversity requires in-depth analysis and understanding through systematic analytical methods [8].

4 | Conclusion

This study used the Napping-UFP sensory method to analyse the similarities and differences in sensory attributes of 18 SABs produced using modern and traditional upper steaming technologies, identifying the cellar aroma differential samples: LZLJ-T7 and LZLJ-M7. The key aroma compounds of LZLJ-T7 and LZLJ-M7 were determined, according to the cAEDA, multiple quantitation methods, calculation of OAVs, and recombination and omission experiments. Combining the results of OPLS-DA and omission experiments, hexyl acetate, ethyl acetate, octyl acetate, ethyl phenylacetate, DL-2-hydroxy-4-methylpentanoate, ethyl 3-phenylpropanoate, butyl hexanoate, pentyl acetate, isoamyl acetate, butyl acetate, *n*-hexanol, isoamyl alcohol, isovaleric acid, and dimethyl trisulfide are identified as the 14 key differential aroma compounds. Using PLSR to analyse the correlation between key differential compounds and aroma attributes in LZLJ-T7 and LZLJ-M7, it was found that hexyl acetate, ethyl 3-phenylpropanoate, isovaleric acid, dimethyl trisulfide, and octyl acetate contribute the most to cellar aroma. Future research can further explore the influence of different upper steaming process parameters on the flavour profile of base Baijiu. This includes optimising and adjusting factors such as steaming pressure and rate, aiming to achieve more precise

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control and improved flavour presentation. Furthermore, the identification of these specific 14 key differential compounds, particularly those highly correlated with cellar aroma, provides crucial molecular targets for future studies. These compounds can serve as valuable aroma markers for scientifically guided optimisation of upper steaming technologies and objective quality control across production batches. Beyond process parameter refinement, these findings offer promising avenues for improving flavour consistency, developing targeted strategies to accentuate desirable traditional sensory notes in modern production, and informing the flavour design of base Baijiu. The methodology and insights gained could also be applied to systematically explore the impact of upper steaming variations in other Baijiu aroma types, contributing significantly to the scientific understanding and advancement of Baijiu fermentation technology.

Author Contributions

Xinran Dai: investigation, visualisation, writing – original draft, writing – review and editing. **Siting Ye:** writing – original draft. **Xiaolong Shang:** writing – review and editing. **Yintao Jia:** writing – original draft, Writing – review and editing. **Baoguo Sun:** supervision. **Caihong Shen:** methodology, resources. **Miao Liu:** project administration, resources. **Feng Lin:** data curation, resources. **Wei Dong:** conceptualisation, writing – review and editing, funding acquisition. **Xiaotao Sun:** conceptualisation, writing-original draft, writing – review and editing, funding acquisition. **Yanfei Xiong:** supervision, resources. **Bo Deng:** supervision, funding acquisition.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors have nothing to report.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Supplementary material for review only. **Data S2:**supplementary information **Figure S1:** Napping-UFP data structure. Data is arranged in 21 blocks: 20 active groups corresponding to the Napping coordinates (values of X and Y in cm) and one illustrative group corresponding to the descriptors from UFP. **Figure S2:** Key differential aroma compounds of LZLJ-T7 and LZLJ-M7. (Venn diagram of the key aroma compounds and differential markers of LZLJ-T7 and LZLJ-M7, A represents differential markers). **Table S1:** Source and purity of qualitative and quantitative compounds. **Table S2:** Quantitative analysis parameters of the important aroma compounds (FD $\geq$ 9) in LZLJ-T and LZLJ-M7. **Table S3:** Comparison of main aroma compounds in LZLJ-T7 and LZLJ-M7.

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